5 **(a)** Define the capacitance of a parallel plate capacitor.

....

....

.... [2]

(b) Two capacitors X and Y have capacitances C_X and C_Y respectively. The capacitors are connected in parallel to a supply that has electromotive force (e.m.f.) E, as shown in Figure 5.1.

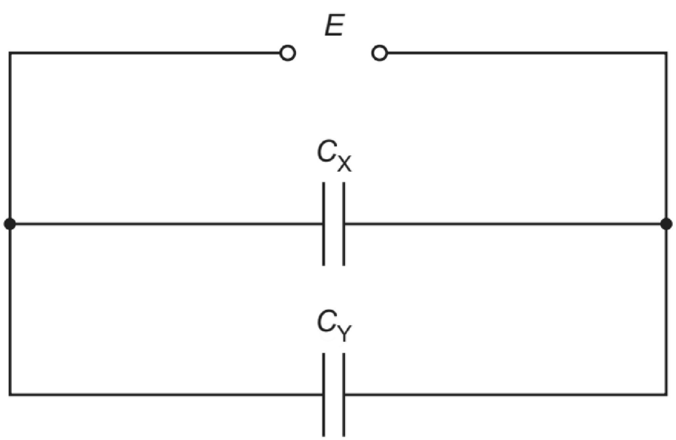

Figure 5.1

Derive the expression for the relationship between C_X, C_Y and the combined capacitance C_T of the circuit.

[3]

(c) Two capacitors are connected in **series** to a supply that has e.m.f. 6.00 V.

One of the capacitors has capacitance C and the other has capacitance $2C$.

The total energy stored by the two capacitors is 1.30 mJ.

Calculate C in μF.

C = ... μF [3]

[Total: 8]